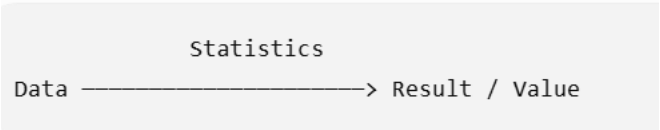
Statistics and Probability

Chapter 1

Exercise 1.1 Statistics Basics

Statistics

Statistics is a science of data in which learn methods or procedure used to collect, present, summarize, analyze and predict that data.



Population Vs Sample

1. Population

In statistics, population means the entire group of people, objects, or things that we want to study.

Example of Population

- Suppose we want to study the **marks of BSE Fall 2023 students registered in MTH262.**
- **Population:** All BSE Fall 2023 students registered in MTH262.

2. Sample

A **sample** means a small group of people or objects selected from a larger population for a study.

Example:
If there are **200 BSE Fall 2023 students registered in MTH262**, and we collect information from **70 students**, then:

- Population = 200 students
- Sample = 70 students

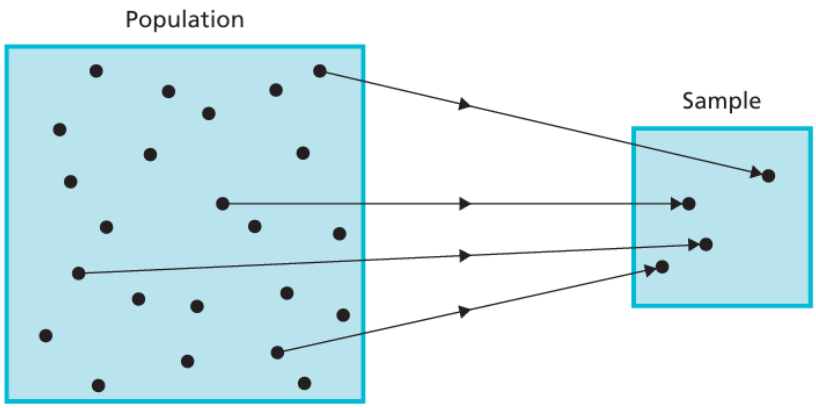
3. What is Sampling?

Sampling is the process of **selecting a smaller group (sample) from a larger group (population)** to collect data and study it.

Example:
If a university has **5,000 students**, we select **200 students** to conduct a survey.

- 5,000 students = Population
- 200 students = Sample
- Selecting the 200 students = Sampling

FIGURE 1.1
Relationship between population and sample



Types of Statistics

1. Descriptive Statistics

Descriptive statistics is the branch of statistics that collects, organizes, summarizes, and presents data so that it is easy to understand.

EXAMPLE 1.1 Descriptive Statistics

The 1948 Baseball Season In 1948, the Washington Senators played 153 games, winning 56 and losing 97. They finished seventh in the American League and were led in hitting by Bud Stewart, whose batting average was .279. Baseball statisticians compiled these and many other statistics by organizing the complete records for each game of the season.

Although fans take baseball statistics for granted, much time and effort is required to gather and organize them. Moreover, without such statistics, baseball would be much harder to follow. For instance, imagine trying to select the best hitter in the American League given only the official score sheets for each game. (More than 600 games were played in 1948; the best hitter was Ted Williams, who led the league with a batting average of .369.)

Example 1.1 (Easy Explanation of Above Example)

In 1948, the Washington Senators played 153 baseball games.

The collected data was:

- **Games played:** 153
- **Games won:** 56
- **Games lost:** 97
- **League position:** 7th
- **Best hitter for the Senators:** Bud Stewart
- **His batting average:** .279

Baseball statisticians collected and organized the records of all the games and summarized them using numbers such as wins, losses, and batting averages.

This is Descriptive Statistics because they are describing and summarizing the data that was collected.

Why is it useful?

Imagine having the score sheet for more than 600 games and trying to find the best hitter manually. It would be difficult.

Statistics makes it easier by summarizing the information. For example:

Ted Williams had a batting average of .369, which was the highest in the American League in 1948.

2. Inferential Statistics

Inferential statistics uses data from a sample to make conclusions or predictions about a larger population.

EXAMPLE 1.2 Inferential Statistics



The 1948 Presidential Election In the fall of 1948, President Truman was concerned about statistics. The Gallup Poll taken just prior to the election predicted that he would win only 44.5% of the vote and be defeated by the Republican nominee, Thomas E. Dewey. But the statisticians had predicted incorrectly. Truman won more than 49% of the vote and, with it, the presidency. The Gallup Organization modified some of its procedures and has correctly predicted the winner ever since.

Political polling provides an example of inferential statistics. Interviewing everyone of voting age in the United States on their voting preferences would be expensive and unrealistic. Statisticians who want to gauge the sentiment of the entire **population** of U.S. voters can afford to interview only a carefully chosen group of a few thousand voters. This group is called a **sample** of the population. Statisticians analyze the information obtained from a sample of the voting population to make inferences (draw conclusions) about the preferences of the entire voting population. Inferential statistics provides methods for drawing such conclusions.

The terminology just introduced in the context of political polling is used in general in statistics.

Example: 1948 Presidential Election (Easy Explanation of Above Example)

In 1948, Harry Truman was running for president against Thomas E. Dewey.

The Gallup Poll wanted to predict who would win. But they could not ask every voter in the USA, because there were too many people.

So, they:

- Selected a small group of voters → Sample
- Asked them who they planned to vote for
- Analyzed their answers
- Used the sample results to make a conclusion about ALL voters → Population

The poll predicted that Truman would get only 44.5% of the vote and would lose.

But the actual result was that Truman won more than 49% and became president.

Why is this Inferential Statistics?

Because they used information from a small sample to make a conclusion about the whole population.

Sample → Analyze data → Make conclusion about Population

More on Descriptive Statistics

EXAMPLE 1.3 Classifying Statistical Studies

The 1948 Presidential Election Table 1.1 displays the voting results for the 1948 presidential election.

TABLE 1.1
Final results of the 1948 presidential election

Ticket	Votes	Percentage
Truman–Barkley (Democratic)	24,179,345	49.7
Dewey–Warren (Republican)	21,991,291	45.2
Thurmond–Wright (States Rights)	1,176,125	2.4
Wallace–Taylor (Progressive)	1,157,326	2.4
Thomas–Smith (Socialist)	139,572	0.3



Exercise 1.7
on page 8

Classification This study is descriptive. It is a summary of the votes cast by U.S. voters in the 1948 presidential election. No inferences are made.

More on Inferential Statistics

EXAMPLE 1.4 Classifying Statistical Studies



Testing Baseballs For the 101 years preceding 1977, the major leagues purchased baseballs from the Spalding Company. In 1977, that company stopped manufacturing major league baseballs, and the major leagues then bought their baseballs from the Rawlings Company.

Early in the 1977 season, pitchers began to complain that the Rawlings ball was "livelier" than the Spalding ball. They claimed it was harder, bounced farther and faster, and gave hitters an unfair advantage. Indeed, in the first 616 games of 1977, 1033 home runs were hit, compared to only 762 home runs hit in the first 616 games of 1976.

Sports Illustrated magazine sponsored a study of the liveliness question and published the results in the article "They're Knocking the Stuffing Out of It" (*Sports Illustrated*, June 13, 1977, pp. 23–27) by L. Keith. In this study, an independent testing company randomly selected 85 baseballs from the current (1977) supplies of various major league teams. It measured the bounce, weight, and hardness of the chosen baseballs and compared these measurements with measurements obtained from similar tests on baseballs used in 1952, 1953, 1961, 1963, 1970, and 1973.

The conclusion was that "...the 1977 Rawlings ball is livelier than the 1976 Spalding, but not as lively as it could be under big league rules, or as the ball has been in the past."

Classification This study is inferential. The independent testing company used a sample of 85 baseballs from the 1977 supplies of major league teams to make an inference about the population of all such baseballs. (An estimated 360,000 baseballs were used by the major leagues in 1977.)

You try it!

Exercise 1.9 on page 8

Shorter Version of Above Case Study

In 1977, pitchers complained that the new Rawlings baseball was "livelier" than the previous Spalding baseball. To investigate, an independent testing company randomly selected 85 baseballs from the 1977 supplies of major league teams and measured their bounce, weight, and hardness. Since it was not practical to test all 360,000 baseballs, the researchers used the sample of 85 to make a conclusion about the larger population. Therefore, this study is Inferential Statistics.

Types of statistical studies

1. Observational Study:

Researchers only observe and collect existing data. They do not change anything or give treatments. It can show an association (relationship) but usually cannot prove causation (cause and effect).

Example: Researchers compare men who already had a vasectomy with men who did not and observe their prostate cancer rates.

EXAMPLE 1.5 An Observational Study

Vasectomies and Prostate Cancer Approximately 450,000 vasectomies are performed each year in the United States. In this surgical procedure for contraception, the tube carrying sperm from the testicles is cut and tied.

Several studies have been conducted to analyze the relationship between vasectomies and prostate cancer. The results of one such study by E. Giovannucci et al. appeared in the paper "A Retrospective Cohort Study of Vasectomy and Prostate Cancer in U.S. Men" (*Journal of the American Medical Association*, Vol. 269(7), pp. 878–882).

Dr. Giovannucci, study leader and epidemiologist at Harvard-affiliated Brigham and Women's Hospital, said that "…we found 113 cases of prostate cancer among 22,000 men who had a vasectomy. This compares to a rate of 70 cases per 22,000 among men who didn't have a vasectomy."

The study shows about a 60% elevated risk of prostate cancer for men who have had a vasectomy, thereby revealing an association between vasectomy and prostate cancer. But does it establish causation: that having a vasectomy causes an increased risk of prostate cancer?

The answer is no, because the study was observational. The researchers simply observed two groups of men, one with vasectomies and the other without. Thus, although an association was established between vasectomy and prostate cancer, the association might be due to other factors (e.g., temperament) that make some men more likely to have vasectomies and also put them at greater risk of prostate cancer.



Exercise 1.19
on page 9

Shorter and Easier Explanation of Above Case Study

This study examined whether there was a relationship between **vasectomy and prostate cancer**. Researchers observed two groups of men: about **22,000 men who had a vasectomy** and men who did not. They found **113 cases of prostate cancer** among men with vasectomies compared with **70 cases per 22,000** among those without. However, the researchers **did not assign or control** who received a vasectomy; they simply observed existing groups. Therefore, it is an **Observational Study**, and it can show an **association**, but it cannot prove that vasectomy **causes** prostate cancer.

Vasectomy and Prostate

Vasectomy is a surgical procedure for male birth control. It involves cutting or blocking the tubes that carry sperm, so sperm cannot enter the semen.

Prostate is a small gland in the male reproductive system, located below the bladder. It helps produce the fluid that carries sperm.

2. Designed Experiment:
Researchers actively give treatments or make changes and then observe the results. This can help determine causation.

Example: Researchers give one group of patients a new medicine and another group a placebo, then compare their results.

EXAMPLE 1.6 A Designed Experiment

Folic Acid and Birth Defects For several years, evidence had been mounting that folic acid reduces major birth defects. Drs. A. E. Czeizel and I. Dudas of the National Institute of Hygiene in Budapest directed a study that provided the strongest evidence to date. Their results were published in the paper "Prevention of the First Occurrence of Neural-Tube Defects by Periconceptional Vitamin Supplementation" (*New England Journal of Medicine*, Vol. 327(26), p. 1832).

For the study, the doctors enrolled 4753 women prior to conception and divided them randomly into two groups. One group took daily multivitamins containing 0.8 mg of folic acid, whereas the other group received only trace elements (minute amounts of copper, manganese, zinc, and vitamin C). A drastic reduction in the rate of major birth defects occurred among the women who took folic acid: 13 per 1000, as compared to 23 per 1000 for those women who did not take folic acid.

In contrast to the observational study considered in Example 1.5, this is a designed experiment and does help establish causation. The researchers did not simply observe two groups of women but, instead, randomly assigned one group to take daily doses of folic acid and the other group to take only trace elements.



Exercise 1.21
on page 9

Shorter and Easier Explanation of Above Case Study

In this study, **4,753 women were randomly divided into two groups**. One group took **folic acid**, while the other received only trace elements. The researchers then compared birth-defect rates: **13 per 1,000** with folic acid versus **23 per 1,000** without it. Because the researchers **gave**

different treatments and randomly assigned the women, this is a **Designed Experiment** and provides evidence of **causation**.

Folic acid

Folic acid is a type of **vitamin B (B9)** that is important for the body.

In pregnant women, it helps the baby's brain and spinal cord develop properly and can reduce the risk of certain birth defects.

Exercise Questions

Understanding the Concepts and Skills

1.1 Define the following terms:
a. Population **b.** Sample

1.2 What are the two major types of statistics? Describe them in detail.

1.4 Explain two ways in which descriptive statistics and inferential statistics are interrelated.

Descriptive and inferential statistics are interrelated in the following two ways:

1. **Descriptive statistics** summarize sample data.
   For example, if we collect the marks of 100 students, we can calculate their average marks. This describes the data of those 100 students.
2. **Inferential statistics** use sample data to draw conclusions about a population.
   The results from those 100 students can be used to make conclusions or estimates about the marks of all students in the university.

1.5 Define the following terms:
a. Observational study **b.** Designed experiment

1.6 Fill in the following blank: Observational studies can re veal only association, whereas designed experiments can help establish

Ans: Causation.

Example

Suppose we want to know whether exercise helps improve exam marks.

- **Observational study:** We observe students and notice that students who exercise more often also tend to get higher marks. This shows an association, but we cannot say exercise is the cause because other factors may be involved.
- **Designed experiment:** We divide students into groups and give one group a planned exercise program while keeping other conditions similar. If the exercise group performs better, the experiment can provide evidence of causation.

In Exercises 1.7–1.12, classify each of the studies as either descriptive or inferential. Explain your answers.

1.7 TV Viewing Times. The Nielsen Company collects and publishes information on the television viewing habits of Americans. Data from a sample of Americans yielded the following estimates of average TV viewing time per month for all Americans 2 years old and older. The times are in hours and minutes (NA, not available). [SOURCE: *Nielsen's Three Screen Report*, May 2008]

Viewing method	May 2008	May 2007	Change (%)
Watching TV in the home	127:15	121:48	4
Watching timeshifted TV	5:50	3:44	56
Using the Internet	26:26	24:16	9
Watching video on Internet	2:19	NA	NA

Answer: Inferential statistics.

Why? The Nielsen Company collected data from a sample of Americans and used it to estimate the TV viewing time of all Americans aged 2 and older.

1.8 Professional Athlete Salaries. In the *Statistical Abstract of the United States*, average professional athletes' salaries in baseball, basketball, and football were compiled and compared for the years 1995 and 2005.

	Average salary ($1000)	
Sport	**1995**	**2005**
Baseball (MLB)	1111	2476
Basketball (NBA)	2027	4038
Football (NFL)	584	1400

Answer: Descriptive statistics.

Why? The study compares the average salaries of professional athletes in baseball, basketball, and football for 1995 and 2005.

It is only summarizing and comparing the available salary data; it is not using a sample to make conclusions about a larger population.

1.9 Geography Performance Assessment. In an article titled "Teaching and Assessing Information Literacy in a Geography Program" (*Journal of Geography*, Vol. 104, No. 1, pp. 17–23), Dr. M. Kimsey and S. Lynn Cameron reported results from an on-line assessment instrument given to senior geography students at one institution of higher learning. The results for level of performance of 22 senior geography majors in 2003 and 29 senior geography majors in 2004 are presented in the following table.

Level of performance	Percent in 2003	Percent in 2004
Met the standard: 36–48 items correct	82%	93%
Passed at the advanced level: 41–48 items correct	50%	59%
Failed: 0–35 items correct	18%	7%

Answer: Descriptive Statistics

Why? The researchers collected and summarized the performance results of 22 students in 2003 and 29 students in 2004 at one institution. They presented the results using percentages.

They are describing the students' performance, not using the sample to make conclusions about a larger population.

1.10 Drug Use. The U.S. Substance Abuse and Mental Health Services Administration collects and publishes data on nonmedical drug use, by type of drug and age group, in *National Survey on Drug Use and Health*. The following table provides data for the years 2002 and 2005. The percentages shown are estimates for the entire nation based on information obtained from a sample (NA, not available).

Type of drug	Percentage, 18–25 years old			
	Ever used		Current user	
	2002	2005	2002	2005
Any illicit drug	59.8	59.2	20.2	20.1
Marijuana and hashish	53.8	52.4	17.3	16.6
Cocaine	15.4	15.1	2.0	2.6
Hallucinogens	24.2	21.0	1.9	1.5
Inhalants	15.7	13.3	0.5	0.5
Any psychotherapeutic	27.7	30.3	5.4	6.3
Alcohol	86.7	85.7	60.5	60.9
"Binge" alcohol use	NA	NA	40.9	41.9
Cigarettes	71.2	67.3	40.8	39.0
Smokeless tobacco	23.7	20.8	4.8	5.1
Cigars	45.6	43.2	11.0	12.0

Answer: Inferential Statistics

Why? The data is collected from a **sample** of people, but the percentages are used to give **estimates for the entire nation**.

They use sample information to make conclusions about the larger population.

1.11 Dow Jones Industrial Averages. The following table provides the closing values of the Dow Jones Industrial Averages as of the end of December for the years 2000–2008. [SOURCE: Global Financial Data]

Year	Closing value
2000	10,786.85
2001	10,021.50
2002	8,341.63
2003	10,453.92
2004	10,783.01
2005	10,717.50
2006	12,463.15
2007	13,264.82
2008	8,776.39

Answer: Descriptive Statistics

Why? The table simply **shows the closing values of the Dow Jones Industrial Average from 2000 to 2008**. It summarizes the recorded data for those years.

It does not use a sample to make conclusions about a larger population.

1.13 Thoughts on Evolution. In an article titled "Who has designs on your student's minds?" (*Nature*, Vol. 434, pp. 1062–1065), author G. Brumfiel postulated that support for Darwinism increases with level of education. The following table provides percentages of U.S. adults, by educational level, who believe that evolution is a scientific theory well supported by evidence.

Education	Percentage
Postgraduate education	65%
College graduate	52%
Some college education	32%
High school or less	20%

a. Do you think that this study is descriptive or inferential? Explain your answer.
b. If, in fact, the study is inferential, identify the sample and population.

a. Answer: Inferential Statistics

Why?
The study uses data from U.S. adults to make a conclusion about the relationship between **education level and support for evolution** among the larger population of U.S. adults.

b. Sample and Population

- **Sample:** The U.S. adults who were included in the study/survey.
- **Population:** All U.S. adults.

1.14 Offshore Drilling. A CNN/Opinion Research Corporation poll of more than 500 U.S. adults, taken in July 2008, revealed that a majority of Americans favor offshore drilling for oil and natural gas; specifically, of those sampled, about 69% were in favor.

a. Identify the population and sample for this study.

b. Is the percentage provided a descriptive statistic or an inferential statistic? Explain your answer.

a. Population and Sample

- **Population**: All U.S. adults.
- **Sample:** The more than 500 U.S. adults who participated in the poll.

b. Answer: Inferential Statistics

Why?
The 69% was calculated from the sample of more than 500 adults and is used to estimate the opinion of all U.S. adults.

Exercise 1.2 Simple Random Sampling

Sampling Unit?

A sampling unit is the individual person, object, or item that can be selected from a population for a sample.

Example

Serial No.	Reg No.	Name	CGPA	Std Hrs	Family Member	Blood Group
1	FA23-BSE-001	Ali	3.2	5	5	A+
2	FA23-BSE-002	Sara	3.5	4	6	B+
3	FA23-BSE-003	Ahmed	3.0	6	4	O+
4	FA23-BSE-004	Hamza	3.7	3	5	A+
5	FA23-BSE-005	Ayesha	3.4	5	7	B-
6	FA23-BSE-006	Usman	2.9	7	6	O+
7	FA23-BSE-007	Hina	3.6	4	4	AB+
...	...	...	...	...	...	...
55	FA23-BSE-055	...	...	...	...	...

Population: All 55 students of FA23-BSE

Sample: The 7 students you selected

Sampling unit: One student

Sampling: The process of selecting those 7 students from the 55 students

Sampling Frame?

A sampling frame is a complete list of all the sampling units in a population from which we select our sample.

Sampling frame: The table/list containing all 55 students with their registration numbers, names, etc.

Census?

A census is a study in which we collect data from every member of the population.

Using your FA23-BSE example:

- **Population:** 55 students
- **Census:** Collect information from all 55 students
- **Sample:** Collect information from only 7 students

Sample Survey?

A sample survey is a survey in which we collect data from only a sample of the population, instead of everyone.

Using your FA23-BSE example:

- **Population:** 55 students
- **Sample:** 7 selected students
- **Sample survey:** Collect information from those 7 students.

Parameter vs Statistic

1. Parameter

A **parameter** is a value calculated from **complete population data**. Because the population is fixed, its parameter is considered **fixed**.

Example:

Suppose a university has **10,000 students**, and we calculate the average CGPA of **all 10,000 students**.

That average is a **parameter**.

If the population doesn't change, the parameter doesn't change.

2. Statistic

A **statistic** is a value calculated from **sample data**. If we take different samples, we can get different results. Therefore, a statistic is considered **variable**.

Example: If we select **7 students** from the 55 and calculate their **average CGPA**, that average is a **statistic**.

Parameter = Population = Fixed

Statistic = Sample = Can change

Experiment?

An **experiment** is a study in which we **change or control something** and observe its effect on another variable.

Example:
Suppose we want to know whether **more study hours improve CGPA**. We give some students a specific study schedule and then compare their CGPA with other students.

Representative Sample?

A **representative sample** is a sample that properly reflects the important characteristics of the population.

Example:
Suppose there are **55 students** in FA23-BSE, with different CGPAs, study hours, and backgrounds. If we select **7 students** in a way that reasonably represents the different types of students in the class, those 7 students form a **representative sample**.

Homogeneity and Heterogeneity

These terms describe **how similar or different the members of a population are**.

1. Homogeneity

Homogeneity means the members are similar to each other.

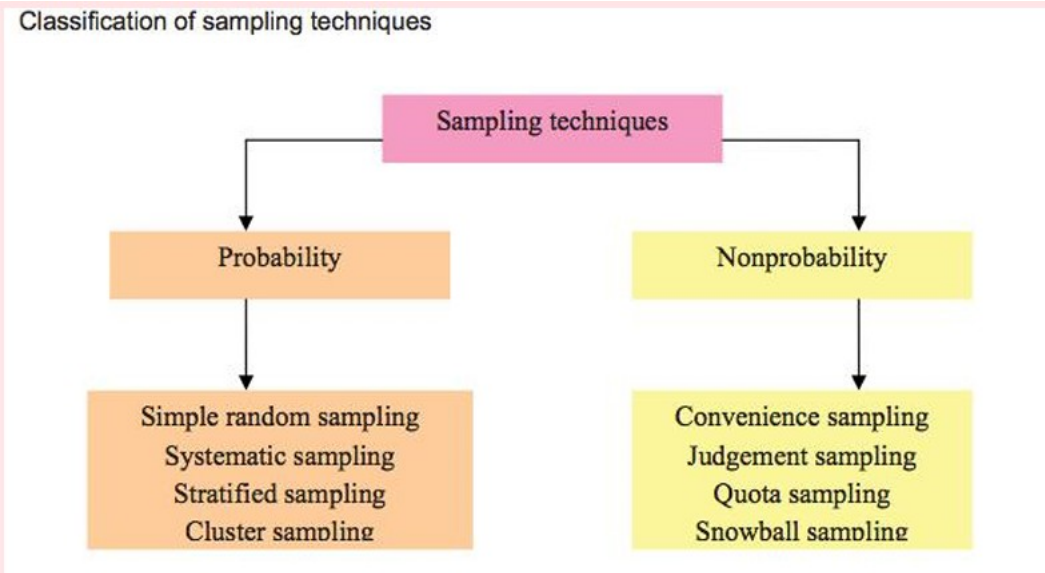
Example:
A class that includes **only male students** is homogeneous because the students have a similar characteristic.

2. Heterogeneity

Heterogeneity means the members are different from each other.

Example:
A class that includes **both male and female students** is heterogeneous because the students have different characteristics.

Sampling Techniques: Probability Sampling vs Non-Probability Sampling



1. Probability Sampling

Every member of the population has a known chance of being selected, usually through random selection.

Example:
There are 55 students in a class. You randomly select 7 students using a lottery or random number generator.

Every student has a chance of being selected.

2. Non-Probability Sampling

Members are selected without random selection, so some members may have no known chance of being selected. The selection may depend on the **selector's choice or preference**, so it can be biased. Therefore, we cannot reliably generalize the results to the whole population.

Example:
A class has **male and female students**, but the selector chooses **only male students** for the sample.

This is **non-random sampling** because the selector chooses who to include.

Types of Probability Sampling

Simple Random Sampling

Every member of the population has an equal and known chance of being selected.

Example:
There are 55 students in a class. We need a sample of 7 students. We write all 55 names on slips and randomly pick 7 slips.

Everyone has an equal chance of being selected.

Simple random sample

A sample obtained by simple random sampling.

EXAMPLE 1.7 Simple Random Samples

TABLE 1.2
Five top Oklahoma state officials

Governor (G)
Lieutenant Governor (L)
Secretary of State (S)
Attorney General (A)
Treasurer (T)

TABLE 1.3
The 10 possible samples of two officials

G, L	G, S	G, A	G, T	L, S
L, A	L, T	S, A	S, T	A, T

TABLE 1.4
The five possible samples of four officials

G, L, S, A	G, L, S, T
G, L, A, T	G, S, A, T
L, S, A, T	



Exercise 1.37 on page 14

Sampling Oklahoma State Officials As reported by the *World Almanac*, the top five state officials of Oklahoma are as shown in Table 1.2. Consider these five officials a population of interest.

a. List the possible samples (without replacement) of two officials from this population of five officials.
b. Describe a method for obtaining a simple random sample of two officials from this population of five officials.
c. For the sampling method described in part (b), what are the chances that any particular sample of two officials will be the one selected?
d. Repeat parts (a)–(c) for samples of size 4.

Solution For convenience, we represent the officials in Table 1.2 by using the letters in parentheses.

a. Table 1.3 lists the 10 possible samples of two officials from this population of five officials.
b. To obtain a simple random sample of size 2, we could write the letters that correspond to the five officials (G, L, S, A, and T) on separate pieces of paper. After placing these five slips of paper in a box and shaking it, we could, while blindfolded, pick two slips of paper.
c. The procedure described in part (b) will provide a simple random sample. Consequently, each of the possible samples of two officials is equally likely to be the one selected. There are 10 possible samples, so the chances are $\frac{1}{10}$ (1 in 10) that any particular sample of two officials will be the one selected.
d. Table 1.4 lists the five possible samples of four officials from this population of five officials. A simple random sampling procedure, such as picking four slips of paper out of a box, gives each of these samples a 1 in 5 chance of being the one selected.

Exercise Set 1.2

1.27 Explain why a census is often not the best way to obtain information about a population.

A census means collecting information from every member of a population.

It is often not the best way because it can be expensive, time-consuming, and difficult to study everyone. A sample can provide information about the population more quickly and at lower cost.

1.28 Identify two methods other than a census for obtaining information.

Two methods other than a census for obtaining information are:

1. Sample survey
2. Experiment

1.29 In sampling, why is obtaining a representative sample important?

A **representative sample** is important because it **accurately reflects the characteristics of the whole population**.

It helps us make **correct conclusions about the population** from the sample.

1.30 Memorial Day Poll. An online poll conducted over one Memorial Day Weekend asked people what they were doing to observe Memorial Day. The choices were: (1) stay home and relax, (2) vacation outdoors over the weekend, or (3) visit a military cemetery. More than 22,000 people participated in the poll, with 86% selecting option 1. Discuss this poll with regard to its suitability.

Answer

This poll is **not suitable for representing the whole population** because it is an **online voluntary poll**. People choose themselves whether to participate, so the sample may be **biased**.

Also, people who were **staying home** were more likely to see and answer the online poll than people who were **outdoors or visiting a cemetery**.

Therefore, the **86% result cannot be generalized to the entire population**.

1.31 Estimating Median Income. Explain why a sample of 30 dentists from Seattle taken to estimate the median income of all Seattle residents is not representative.

This sample is **not representative** because it includes only **dentists**, while the population includes **all Seattle residents**. Dentists may have different incomes from other residents.

Therefore, the sample **does not properly represent the whole population**.

1.32 Provide a scenario of your own in which a sample is not representative.

Answer

A teacher wants to know the **average marks of all students in a class**, but selects only **10 students who got the highest marks**.

This sample is **not representative** because it does not include students with average or low marks.

1.33 Regarding probability sampling:

a. What is it?
b. Does probability sampling always yield a representative sample? Explain your answer.
c. Identify some advantages of probability sampling.

a. What is probability sampling?
Every member of the population has a known chance of being selected, usually through random selection.

b. Does it always give a representative sample?
No. Probability sampling does **not always** give a representative sample. A sample can be selected randomly but still, by chance, **not perfectly represent the population**.

c. Advantages of probability sampling:

- It **reduces bias** in selecting the sample.
- Every member has a **known chance of selection**.
- It allows us to **generalize results** from the sample to the population.
- It is more **reliable and objective** than non-probability sampling.

1.34 Regarding simple random sampling:

a. What is simple random sampling?
b. What is a simple random sample?
c. Identify two forms of simple random sampling and explain the difference between the two.

a. What is simple random sampling?

Every member of the population has an equal and known chance of being selected.

b. What is a simple random sample?

A sample obtained by simple random sampling.

c. Two forms of simple random sampling:

1. **Sampling with replacement:**
   After selecting a member, we put them back before selecting again. So, the same member can be selected more than once.
2. **Sampling without replacement:**
   After selecting a member, we do not put them back. So, the same member cannot be selected again in the same sample.

1.36 Identify two methods for obtaining a simple random sample.

Two methods are:

1. **Lottery method:** Write the names/numbers of all population members on separate pieces of paper, mix them, and randomly select the required number.
2. **Random number method:** Assign a number to each population member and use a random number generator/table to select the sample.

1.37 Question

Oklahoma State Officials. The five top Oklahoma state officials are displayed in Table 1.2 on page 12. Use that table to solve the following problems.

a. List the 10 possible samples (without replacement) of size 3 that can be obtained from the population of five officials.

b. If a simple random sampling procedure is used to obtain a sample of three officials, what are the chances that it is the first sample?

Answer

The five officials are:

G, L, S, A, T

a. 10 possible samples of size 3

1. G, L, S
2. G, L, A
3. G, L, T
4. G, S, A
5. G, S, T
6. G, A, T
7. L, S, A
8. L, S, T
9. L, A, T
10. S, A, T

b. Chance of selecting the first sample

There are 10 possible samples.

Since simple random sampling gives every sample an equal chance:

So, the chance of selecting the first sample (G, L, S) is:

1 out of 10 = 10%.

1.38 Best-Selling Albums

The Recording Industry Association of America provides data on the best-selling albums of all time. As of January 2008, the top six best-selling albums of all time (U.S. sales only) are by the artists **Eagles (E),**

Michael Jackson (M), Pink Floyd (P), Led Zeppelin (L), AC/DC (A), and Billy Joel (B).

a. List the 15 possible samples (without replacement) of two artists that can be selected from the six. For brevity, use the initial provided.

b. Describe a procedure for taking a simple random sample of two artists from the six.

c. If a simple random sampling procedure is used to obtain two artists, what are the chances of selecting **P and A? M and E?**

Answer

a. 15 possible samples of two

The six artists are: **E, M, P, L, A, B**

1. E, M
2. E, P
3. E, L
4. E, A
5. E, B
6. M, P
7. M, L
8. M, A
9. M, B
10. P, L
11. P, A
12. P, B
13. L, A
14. L, B

15. A, B

So, there are **15 possible samples**.

b. Simple random sampling procedure

Write the six initials (**E, M, P, L, A, B**) on separate pieces of paper. Put them in a box, mix them, and **randomly pick two papers**.

The two selected artists form the **simple random sample**.

c. Chances of selecting P and A / M and E

There are **15 possible samples**, and each has an equal chance.

P and A:

M and E:

So, **both samples have a 1 in 15 chance**, or approximately **6.67%**.

1.3 Other Sampling Designs

2. Systematic Random Sampling

Systematic Random Sampling is a probability sampling method where we select members at a regular interval after choosing a random starting point.

Method

Instead of randomly choosing every person separately, we:

1. Find the **interval** .
2. Randomly choose a **starting number** .
3. Then select every -th person.

Example

Suppose we have **100 students** and need a sample of **10**.

Step 1:
So, **m = 10**

Step 2:
Randomly choose a number from **1 to 10**.
Suppose we get **4**.

Step 3:
Start from 4 and select every 10th student:

4, 14, 24, 34, 44, 54, 64, 74, 84, 94

These 10 students are our **systematic random sample**.

2. Cluster Sampling

Cluster sampling is a probability sampling method in which the population is divided into groups called clusters, and some clusters are randomly selected for the sample.

Example:
A university has 20 classes. Randomly select 5 classes and collect data from all students in those selected classes.

Remember: Select some whole groups.

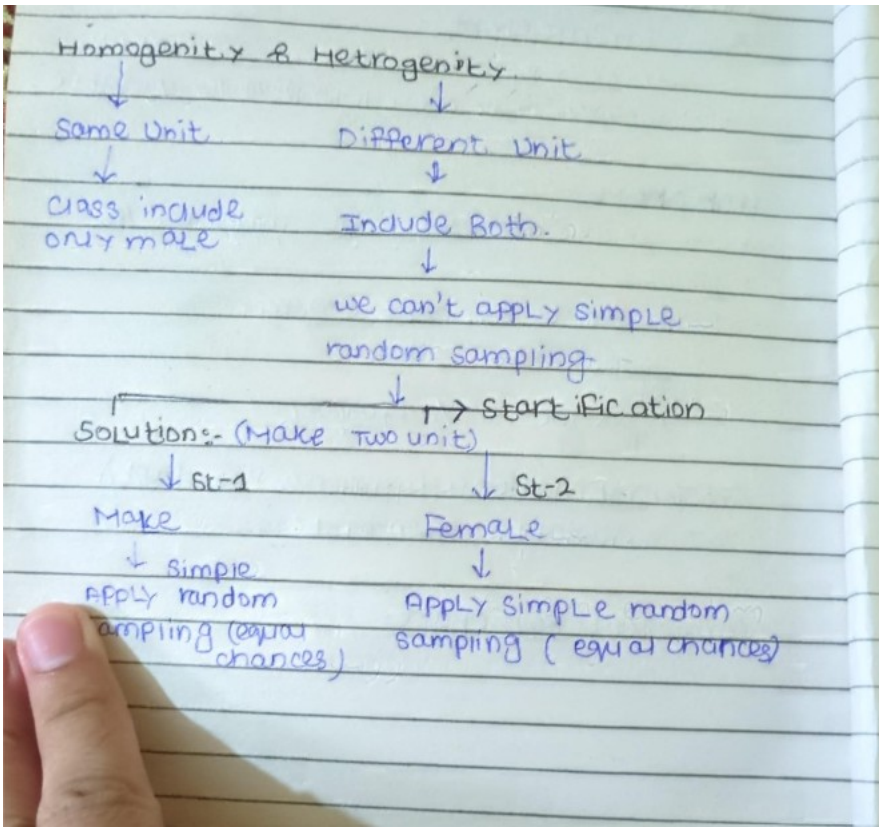
3. Stratified Sampling

Stratified sampling is a probability sampling method in which the population is divided into different groups (strata), and some members are randomly selected from each group.

Example:
A university has 600 male and 400 female students. To select 100 students, randomly select students from both male and female groups.

Remember: Some members from every group.

Why use stratified sampling instead of simple random sampling in case of Homogeneity?



Suppose a university has male and female students:

- Population = 70% male, 30% female
- If we use simple random sampling, we randomly select students from the whole population.
- By chance, we might select more males and very few females.
- So, one group may not be properly represented.

Therefore, we divide the population into two homogeneous strata:

Stratum 1 → Male → apply simple random sampling
Stratum 2 → Female → apply simple random sampling

This is called stratified random sampling.

Key Note

When a population contains different groups, simple random sampling may not ensure proper representation of every group. Therefore, we divide the population into homogeneous strata and apply simple random sampling within each stratum. This is called stratified random sampling.

Exercise set 1.3

1.49 The International 500. In Exercise 1.43 on page 15, you used simple random sampling to obtain a sample of 10 firms from *Fortune Magazine*'s list of "The International 500."

a. Use systematic random sampling to accomplish that same task.

b. Which method is easier: simple random sampling or systematic random sampling?

c. Does it seem reasonable to use systematic random sampling to obtain a representative sample? Explain your answer.

1.49 The International 500

There are 500 firms, and we need a sample of 10 firms.

a. Use systematic random sampling

Use the textbook's systematic sampling procedure: divide population size by sample size, randomly choose a starting number, then select every -th member.

Step 1: Find

Step 2: Randomly select a number from 1 to 50.

Suppose the random starting number is 22.

Step 3: Select every 50th firm:

So, these 10 firms form the systematic random sample.

b. Which method is easier?

Systematic random sampling is easier.

Because after selecting the random starting number, we simply select every 50th firm. The textbook also states that systematic random sampling takes less effort to implement than simple random sampling.

c. Is it reasonable for obtaining a representative sample?

Yes, generally.

Systematic random sampling usually provides results comparable to simple random sampling. However, it may not work well if the list has a cyclical pattern (repeating pattern).

1.50 Keno. In the game of keno, 20 balls are selected at random from 80 balls numbered 1–80. In Exercise 1.44 on page 15, you used simple random sampling to simulate one game of keno.

a. Use systematic random sampling to obtain a sample of 20 of the 80 balls.

b. Which method is easier: simple random sampling or systematic random sampling?

c. Does it seem reasonable to use systematic random sampling to simulate one game of keno? Explain your answer.

a. Use systematic random sampling

Step 1: Find

Step 2: Randomly select a starting number from 1 to 4.

Suppose we randomly select 3.

Step 3: Select every 4th number:

These are the 20 balls selected.

b. Which method is easier?

Systematic random sampling is easier.

After choosing the starting number, we just select every 4th ball.

c. Is systematic random sampling reasonable for simulating keno?

No, not really.

In keno, balls are supposed to be selected at random. Systematic sampling forces the numbers to follow a fixed pattern, such as every 4th number. Therefore, it does not realistically simulate the random selection of a keno game.